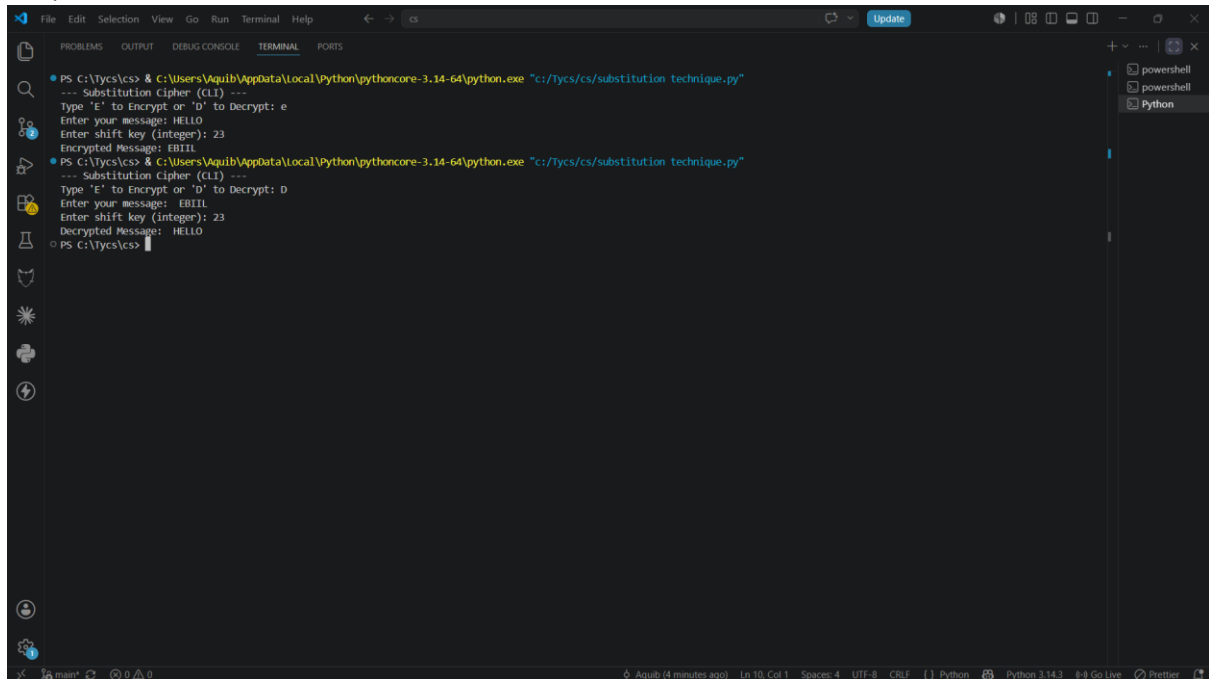


## 1.A

Design and implement algorithms to encrypt and decrypt messages using classical substitution techniques

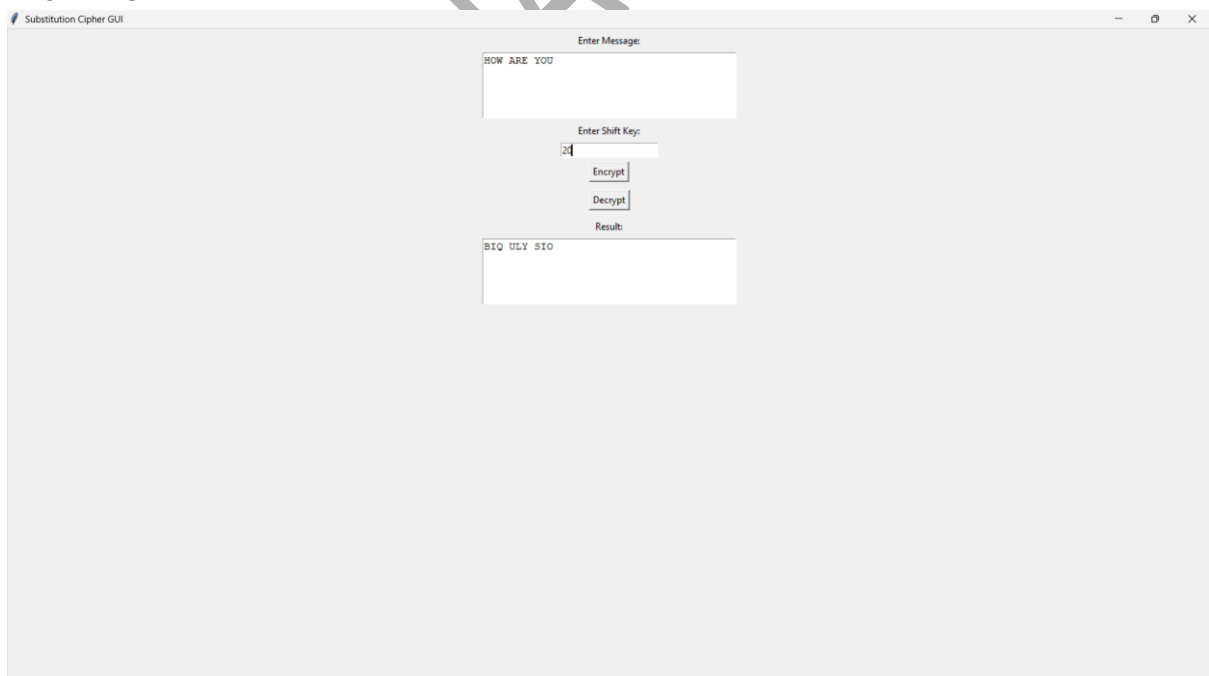
Output: CLI METHOD



```
PS C:\Tycs\cs> & C:\Users\Aqib\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Tycs/cs/substitution technique.py"
--- Substitution Cipher (CLI) ---
Type 'E' to Encrypt or 'D' to Decrypt: e
Enter your message: HELLO
Enter shift key (integer): 23
Encrypted Message: EBILL
PS C:\Tycs\cs> & C:\Users\Aqib\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Tycs/cs/substitution technique.py"
--- Substitution Cipher (CLI) ---
Type 'E' to Encrypt or 'D' to Decrypt: D
Enter your message: EBILL
Enter shift key (integer): 23
Decrypted Message: HELLO
PS C:\Tycs\cs>
```

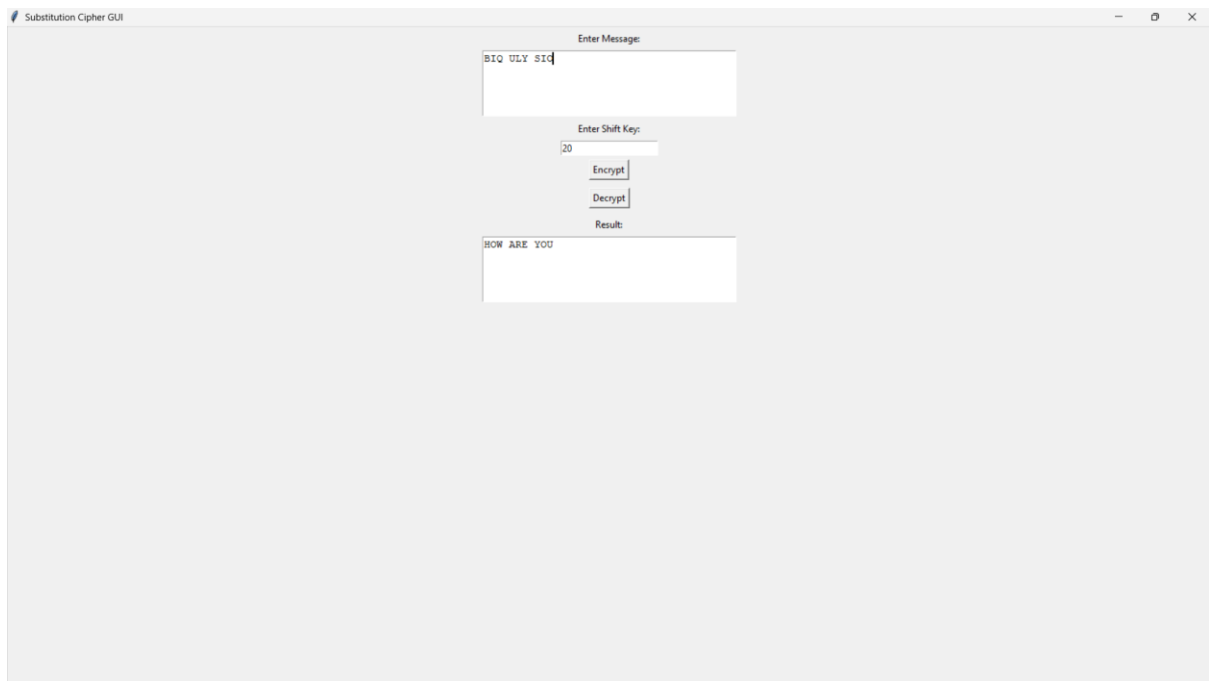
OUTPUT: GUI METHOD

ENCRYPTION:



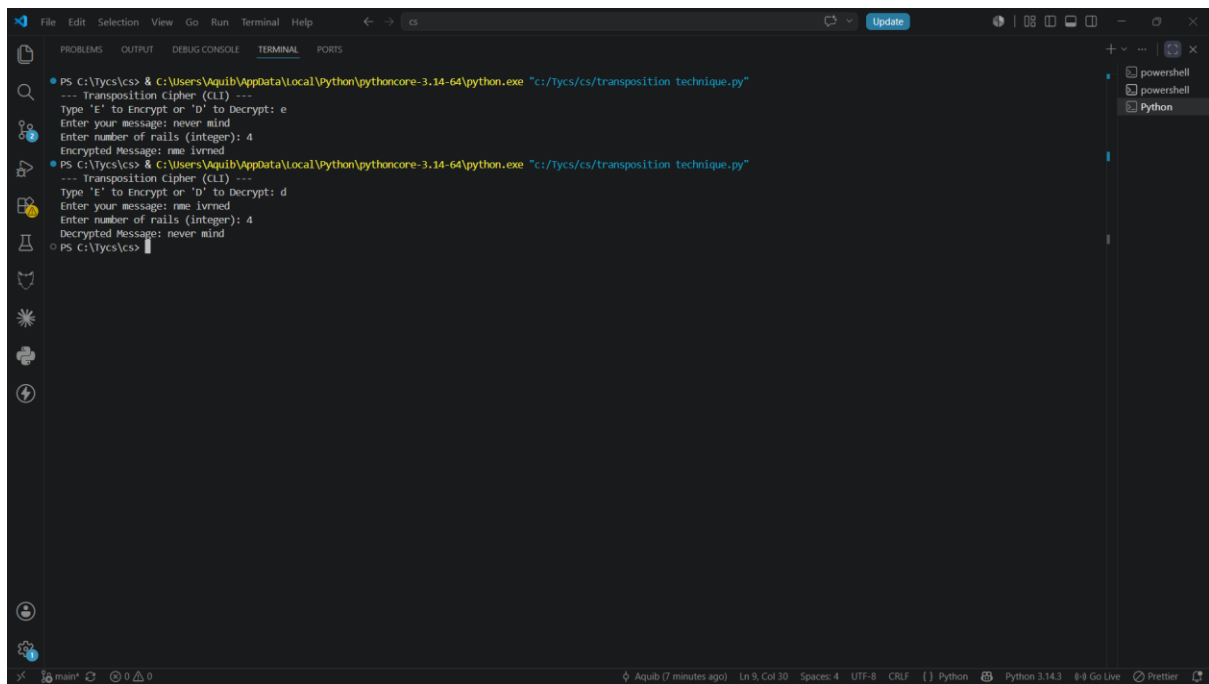
MVLU

DECRYPTION:



The screenshot shows a window titled "Substitution Cipher GUI". Inside, there are two input fields at the top: "Enter Message:" containing "SIQ ULV SIQ" and "Enter Shift Key:" containing "20". Below these are two buttons: "Encrypt" and "Decrypt". The "Decrypt" button is highlighted. At the bottom, a "Result:" label is followed by a text box containing the decrypted message "HOW ARE YOU".

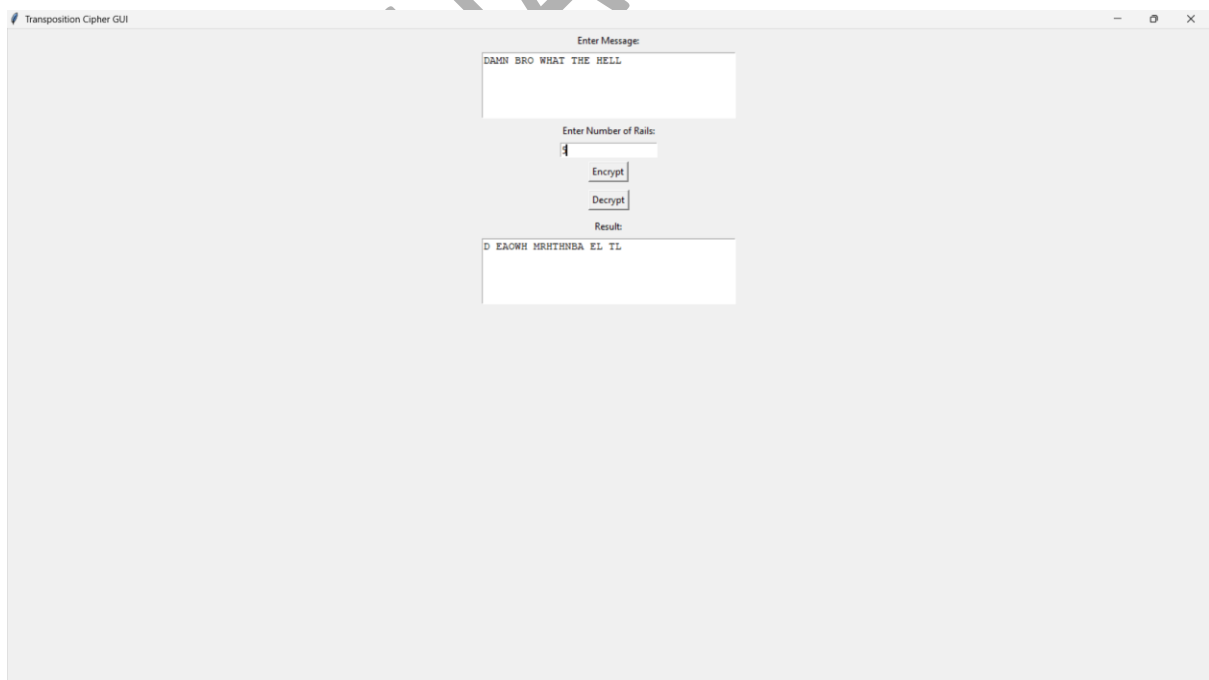
Design and implement algorithms to encrypt and decrypt messages using transposition techniques  
OUTPUT:



```
PS C:\Iyycs\cs> & C:\Users\Aqib\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Iyycs/cs/transposition technique.py"
--- Transposition Cipher (CLI) ---
Type 'E' to Encrypt or 'D' to Decrypt: e
Enter your message: never mind
Enter number of rails (integer): 4
Encrypted Message: nme ivrned
PS C:\Iyycs\cs> & C:\Users\Aqib\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Iyycs/cs/transposition technique.py"
--- Transposition Cipher (CLI) ---
Type 'E' to Encrypt or 'D' to Decrypt: d
Enter your message: nme ivrned
Enter number of rails (integer): 4
Decrypted Message: never mind
PS C:\Iyycs\cs>
```

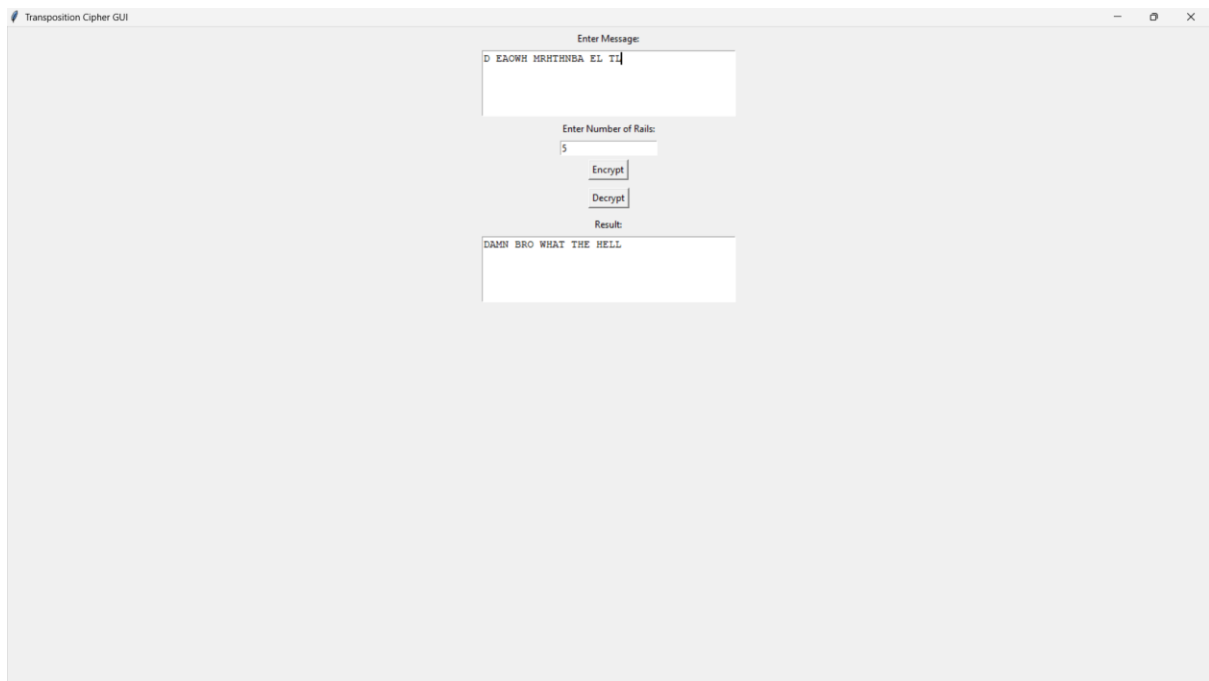
OUTPUT GUI:

ENCRYPTION::



MVLU

DECRYPTION::



The screenshot shows a window titled "Transposition Cipher GUI". Inside, there are two input fields. The first, labeled "Enter Message:", contains the text "D EACWH MRHTHBA EL TI". The second, labeled "Enter Number of Rails:", contains the number "5". Below these fields are two buttons: "Encrypt" and "Decrypt". The "Decrypt" button is highlighted. Below the buttons is a "Result:" label and a text box containing the decrypted message "DAMN BRO WHAT THE HELL".